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# Process evaluation of an 865 $MW_e$ lignite fired $O_2/CO_2$ power plant

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**Abstract**

In order to reduce emissions of carbon dioxide from large point sources, new technologies can be used in capture plants for combustion of fossil fuel for subsequent capture and storage of $CO_2$. One such technology is the $O_2/CO_2$ combustion process (also termed oxy-fuel combustion) that combines a conventional combustion process with a cryogenic air separation process so that the fuel is burned in oxygen and recycled flue gas, yielding a high concentration of $CO_2$ in the flue gas, which reduces the cost for its capture. In this work, the $O_2/CO_2$ process is applied using commercial data from an 865 $MW_e$ lignite fired reference power plant and large air separation units (ASU). A detailed design of the flue gas treatment pass, integrated in the overall process layout, is proposed. The essential components and energy streams of the two processes have been investigated in order to evaluate the possibilities for process integration and to determine the net efficiency of the capture plant. The electricity generation cost and the associated avoidance cost for the capture plant have been determined and compared to the reference plant with investment costs obtained directly from industry. Although an existing reference power plant forms the basis of the work, the study is directed towards a new state of the art lignite fired $O_2/CO_2$ power plant. The boiler power of the $O_2/CO_2$ plant has been increased to keep the net output of the capture and the reference plant similar. With the integration possibilities identified, the net efficiency becomes 33.5%, which should be compared to 42.6% in the reference plant. With a lignite price of 5.2 $/MWh and an interest rate of 10%, the electricity generation cost increases from 42.1 to 64.3 $/MWh, which corresponds to a $CO_2$ avoidance cost of 26 $/ton $CO_2$.


*Keywords:* $O_2/CO_2$ recycle combustion; Oxy-fuel; Process layout; Cost analysis

## 1. Introduction

Capture and storage of $CO_2$ has the potential to contribute to a significant and relatively quick reduction in $CO_2$ emissions from power generation, allowing fossil fuels to be used as a bridge to a non-fossil future and taking advantage of the existing power plant infrastructure. Commonly studied processes in the literature for

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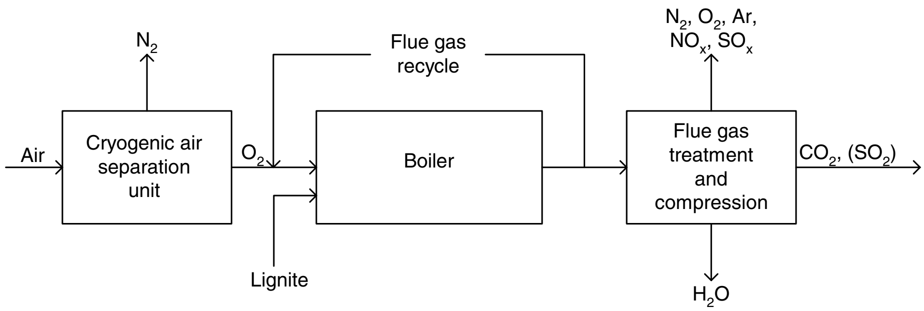



Fig. 1. Principal outline of the $O_2/CO_2$ process.

the purpose of $CO_2$ capture from coal fired power plants are the amine based absorption processes, the integrated gasification combined cycle (IGCC) and the $O_2/CO_2$ combustion process (or oxy-fuel combustion). Plant economics and performance have been evaluated in a number of studies, e.g. [1–6], where the avoidance cost of the captured $CO_2$ is normally used for comparing the costs of the various schemes. This work presents the costs for $CO_2$ capture using the $O_2/CO_2$ combustion process, which is applicable to different types of fuels and boilers. $O_2/CO_2$ combustion involves burning the fuel in an atmosphere of oxygen and recycled flue gas instead of in air, as schematically outlined in Fig. 1. The mixed flow of oxygen and recycled flue gas is fed to the boiler together with the fuel, which is burned as in a conventional plant. Typically, 70–80% of the flue gas is recycled from downstream of the economizer and mixed with the oxygen. The remaining part of the flue gas is cleaned, compressed and later transported to storage or to another application. Studies on $O_2/CO_2$ combustion have mainly concerned emissions and combustion behaviour (e.g. [7,8]) together with overall process studies (e.g. [6,1]). This work combines a comprehensive study of flue gas treatment together with the integration possibilities of the $O_2/CO_2$ process, resulting in a proposal for an overall process layout. Commercial process data are applied in order to identify the possible problems of the components in the process and to obtain design requirements under conditions that are as real as possible. Both absorption based capture processes (e.g. MEA absorption) and $O_2/CO_2$ combustion are often considered as alternatives for retro-fitting existing coal fired units [1,2], making it possible to take advantage of the already invested capital, which, to a certain extent, can be considered as sunk costs. However, existing units are often old units with rather modest net efficiencies, and with the increased parasitic losses of the capture, the retro-fit concept results in comparatively high fuel costs. It should, therefore, be pointed out that although an existing reference power plant has been the basis of the process design of this study, the present work should be considered as a feasibility study for a new $O_2/CO_2$ fired power plant, including costs, process integration and optimization of the steam cycle, ASU (air separation unit) and flue gas treatment pass and applying commercial state of the art process and costs data as the basis. Furthermore, as a comparison, the associated costs are determined for a capture plant both with and without an $SO_2$ removal system (wet flue gas desulphurization (FGD)) in order to show the possible economic benefit from combined capture of $SO_2$ and $CO_2$, i.e. in case this will be environmentally approved and applicable to the type of storage considered.

## 2. Method

The $2 \times 865$ MW lignite fired Lippendorf power plant is used as the reference in this study. This is a modern state of the art power plant that was commissioned in the year 2000. Table 1 lists the main process data of the plant. In [9], more detailed information is available on the process integration part of the study. Further details are also given in [10,11]. Since the focus in the previous work was rather on the performance and design of the $O_2/CO_2$ process, the plant was derated compared to the reference plant due to the increased internal electricity demand of the equipment added, i.e. mainly the ASU and the $CO_2$ compressors. Since the aim of this study is to illustrate the environmental implications and costs associated with a new $O_2/CO_2$ power plant, the net electricity output is kept the same as in the reference plant. This results in an increased boiler power of the capture plant, which is obtained by multiplying the reference boiler power with a simple scaling factor:

Table 1
Main process data for the reference power plant Lippendorf (same data for both blocks)

| | |
|---|---|
| Gross electricity output | 933 MW |
| Net electricity output | 865 $MW_e$ |
| Boiler power | 2030 MW |
| District heat extraction | 115 MW |
| Electricity net efficiency | 42.6% |
| Fuel | Raw lignite |
| Steam flow | 672 kg/s |
| High pressure steam | 554/258.5 °C/bar |
| Intermediate steam pressure | 583/49.7 °C/ bar |
| Steam pressure at condenser discharge | 0.038 bar |

$$\mathrm{Sf} = \frac{\eta_{\mathrm{ref}}}{\eta_{\mathrm{O_2/CO_2}}} \tag{1}$$

where $\eta_{\mathrm{ref}}$ and $\eta_{\mathrm{O_2/CO2}}$ represent the net efficiencies of the reference plant and the $O_2/CO_2$ plant, respectively. Since Sf is not far from 1 (1.3 with the efficiencies of this work), linear scaling is considered a reasonable approximation. The fractional reduction in energy output, or energy penalty (EP), is related to the same reduction in net efficiency of the capture plant and is determined according to

$$\mathrm{EP} = 1 - \frac{\eta_{\mathrm{O_2/CO_2}}}{\eta_{\mathrm{ref}}} \tag{2}$$

What may appear as a base case recycle mixture with 21 vol.% $O_2$ and 79% flue gas will probably not correspond to a real design case. In fact, experimental studies on both gas and coal fired $O_2/CO_2$ combustion, see e.g. [12,13], show that air like combustion conditions are rather obtained for higher $O_2$ concentrations of about 30 vol.%, which are obtained by means of a reduced recycle rate. Ultimately, such a reduction would lead to a certain reduction of both the size and cost of the boiler house. This is, however, kept outside the scope of this work, and for simplicity, similar volumetric flow conditions have been kept in the $O_2/CO_2$ and the reference plant, where the recycled feed gas consists of about 21 vol.% oxygen and 79 vol.% flue gas. This facilitates a direct comparison with the reference plant with respect to equipment and flows, and hence, for the $O_2/CO_2$ concept studied, the plant design before flue gas recycling is assumed identical with the reference plant if not otherwise stated.

The process data and schemes were obtained directly from the plant owner (Vattenfall). Based on these data, a process evaluation was performed in order to identify the new components needed as well as the components that can be excluded from the $O_2/CO_2$ scheme. The process layout is based on an existing ASU with an $O_2$ production of 50,000 $m_n^3/h$. Different compressor configurations for the ASU, as well as for the flue gas compression, were analysed, and calculations of the adiabatic compressor work for different compressor cycle designs were performed using the software RefCalc [14] from which pressure enthalpy diagrams and their corresponding numeric values are obtained. The properties of the various gas mixtures are obtained from the NIST standard reference database with the software Refprop [14] (Reference Fluid, Thermodynamic and Transport Properties).Two different chemical process simulation programs, ChemCad (v5.0) [15] and Hysys (v4.2) [16] were used to simulate the chemical reactions in the flue gas condenser for cross comparison of results. ChemCad, which, in this case, is the most accurate one, uses electrolyte reactions together with thermodynamic models (Peng-Robinson). Hysys does not consider the electrolyte reactions, and hence, for the dissolved compounds, the accuracy is lower than in the Chemcad program. Hysys was also used to determine the total heat rejection and the temperature curve of the condensation unit. In the proposed scheme, $NO_x$ is separated in a liquid/gas separator, since it is assumed not to be soluble in the liquid $CO_2/SO_2$ mixture.

The actual fuel flow and coal composition at the Lippendorf plant is given in [11] for three different coal qualities, which are listed in Table 2: a guarantee coal and a max water and a max ash coal. In a similar way, as in [11], the guarantee coal is used for all calculations regarding efficiency and flue gas flows. The plant must, nevertheless, be able to handle other coal as well. Thus, the so called "Max ash" and "Max water" coals are used for dimensioning the flue gas cleaning equipment. The gas flows, presented in Table 3, are

Table 2
Proximate analysis [kg/kg] and fuel flow [kg/s] of the lignite

| | Guarantee | Max water | Max ash |
|---|---|---|---|
| Hi [MJ/kg] | 10.5 | 9.7 | 9.7 |
| C | 0.2911 | 0.2917 | 0.2718 |
| H | 0.02470 | 0.0248 | 0.0231 |
| O | 0.0819 | 0.0821 | 0.0765 |
| N | 0.0030 | 0.0030 | 0.0028 |
| S | 0.0143 | 0.0143 | 0.0136 |
| Cl | 0.00010 | 0.00010 | 0.00010 |
| F | 0.00005 | 0.00005 | 0.00005 |
| Ash | 0.0650 | 0.0541 | 0.0850 |
| Moisture | 0.5200 | 0.5300 | 0.5276 |
| Fuel flow | 192.6 | 223.7 | 223.7 |

Table 3
Design composition of the flue gas during $O_2/CO_2$ combustion

| Components | [kg/s] | [wt%] | [$m_n^3/s$] | [vol%] |
|---|---|---|---|---|
| $H_2O$ | 176.4 | 38.4 | 222.3 | 60.4 |
| $CO_2$ | 253.9 | 55.3 | 130.2 | 35.4 |
| $SO_2$ | 6.7 | 1.5 | 2.3 | 0.6 |
| $O_2$ | 6.4 | 1.4 | 4.6 | 1.2 |
| $N_2$ | 0.6 | 0.2 | 0.5 | 0.2 |
| Ar | 14.6 | 3.2 | 8.3 | 2.2 |
| Total | 458.7 | 100 | 365.9 | 100 |

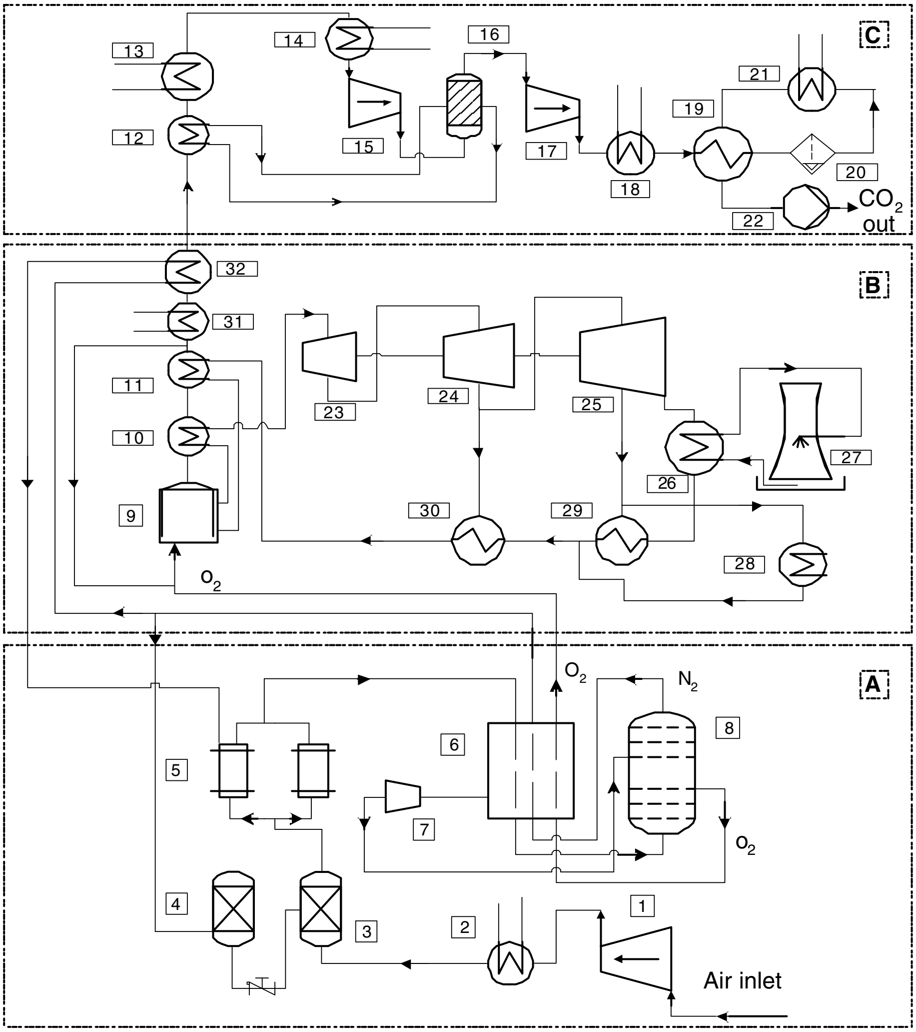
calculated based on the coal analysis given in Table 2. The air excess in the reference boiler is 15% corresponding to an excess of 3% $O_2$. It is, however, likely that the oxygen excess could be further decreased on a full scale $O_2/CO_2$ plant due to the flue gas recycle, and therefore, a value of 1.5% $O_2$ (dry basis) excess is used in this study.

All investment costs have been determined in co-operation with industry [17–19]. The specific investment cost and variable operation and maintenance cost of the power plant with capture has been assumed equal to that of the reference power plant. In the case where the FGD is included, the investment cost has been reduced to 60% of the corresponding FGD for the reference plant [19], since the flue gas flow will decrease drastically in the $O_2/CO_2$ scheme. According to the literature, see e.g. [20], wet FGD technology can be applied, although the $SO_2$ removal is to be performed in a $CO_2$ atmosphere with 3 times as high volumetric concentrations of $SO_2$ as under normal conditions in air firing. Compared to what has been previously reported by the authors [10], the investment and running costs of the ASU has been up dated for the increased plant size. These costs were obtained from [17] for three different unit sizes with a production capacity ranging from 276,900 to 528,700 $m_n^3/h$ of $O_2$. The specific power consumption of these ASU units ranges from 0.34 to 0.36 $kW/m_n^3/h$ according to [17] for a unit producing $O_2$ of 99.6% purity. The power consumption and the investment costs are reduced when a lower purity is required. In this study, all the calculations are made for an $O_2$ purity of 95% and the reduction in power demand is about 1.6% and 4% for the investment costs based on the information given in [10].

A cost analysis for each process part (Power plant, ASU and flue gas treatment pass) was set up in order to obtain the electricity generation cost and the avoidance cost. The electricity generation cost is the total annualized cost (€/year) divided by the annual electricity production (kWh/year). The avoidance cost (€/ton $CO_2$) is calculated according to

$$(\text{€}/\text{ton CO}_2) = \frac{(\text{€}/\text{MWh})_{\text{w}} - (\text{€}/\text{MWh})_{\text{w/o}}}{(t\text{CO}_2/\text{MWh}_{\text{w/o}} - t\text{CO}_2/\text{MWh}_{\text{w}})} \tag{3}$$

where the indices ‘w’ and ‘w/o’ denote a power plant with and without $CO_2$ capture, respectively ($/€ = 1.30 according to OECD [21]). As described above, this calculation refers to plants with and without capture with the same net electric output, which is an important constraint both from an energy systems perspective and when discussing the economic and environmental implications of the capture plant.




1. Air compressor
2. Compressor cooling
3. Direct contact air cooler
4. Evaporative cooler
5. Molecular sieves
6. Heat exchanger
7. Expansion turbine
8. Destillation column
9. Boiler
10. Super heater
11. Economizer
12. TEG heat exchanger
13. Flue gas condensation unit
14. Flue gas cooler or FGD unit
15. Compressor unit 1, 30 bar
16. TEG
17. Compressor unit 2, 58 bar
18. $CO_2$ condenser
19. Heat exchanger ($CO_2$/$CO_2$)
20. Gas/Liquid separator
21. Subcooler
22. High pressure pump
23. HP Steam turbine
24. IP Steam turbine
25. LP Steam turbine
26. Condenser
27. Cooling tower
28. District heating
29. Feed water preheater (FPH)
30. FPH
31. Optional heater, district heat/FPH
32. Nitrogen heater

Fig. 2. Overall process layout for the $O_2/CO_2$ plant proposed in this work. The plant scheme is made with the Lippendorf plant as reference: (A) air separation unit, (B) power plant, (C) flue gas treatment pass.

## 3. Results

### 3.1. Process evaluation

Fig. 2 gives a principal process scheme of the lignite fired $O_2/CO_2$ plant as obtained from the process evaluation using Lippendorf as the reference plant. The three main parts of the plant are the ASU (A), the power boiler island (B) and the flue gas treatment pass (C) with the essential features and components described below following the numbering of Fig. 2.

The ASU is based on cryogenic air separation, which is the only separation technique that can provide the oxygen flows required in the present application [22]. An oxygen production rate of 528,000 $m_n^3/h$ is required and can be split into either 2 or 4 ASU production lines with a corresponding maximum production capacity per line of 277,000 or 138,000 $m_n^3/h$ [17]. An oxygen purity of 95% is selected as the most favourable, since it gives an exchange rate (oxygen to oxygen) of 1.0 without nitrogen in the product gas, but with nearly 5% argon content. Thus, for oxygen purities lower than 95%, the oxygen contains nitrogen in addition to the impurity of argon. The final power consumption of the ASU with the above features becomes 181 $MW_e$. The compressor(s) (1) in the ASU, with intercooling in four steps, operates between ambient temperature and about 60 °C. Without intercooling, an air temperature of about 210 °C is reached, with a heat rejection of about 1 $MW_t$ per $MW_e$ consumed, which could be used for feed water preheating or district heating. However, this would result in a significant decrease in compressor efficiency of approximately 20%, which makes this alternative unattractive. The cooling of both the $CO_2$ compressors (15) and (17) and the air compressor (1) is performed with cooling water from the plant cooling tower (27). In addition, the $CO_2$ condensation also requires cooling water from the cooling tower. In total, there must be an almost 50% increase in the mass flow of cooling water produced in the cooling towers compared to the reference plant in order to attain the low temperatures for the compressor intercooling and the $CO_2$ condensation. However, if the temperature levels are increased only a few degrees, the coolant mass flow will decrease significantly due to the narrow temperature intervals set for cooling the components ($t_{in} = 16$ °C, $t_{in} = 22$ °C on a year average). To minimize losses in power transmission to the air and $CO_2$ compressors, these can be driven directly by steam turbines on a joint shaft. However, the main steam turbine shaft cannot be used for this purpose, since it would cause problems both in the compressor units, such as surge at start up and in the shaft itself because of too large thermal motions. The extra investment cost for the separate powering of the compressors is to be compared with the power saving, corresponding to about 0.7% of the net output. The heat required in the molecular sieves (5) is provided by the nitrogen heater (32), which exchanges heat from the flue gas with a minimum temperature of 200 °C. A cooling capacity of about 25 MW at a temperature of 8 °C can be generated in the evaporative cooler (4) that can be used in the flue gas treatment for subcooling the carbon dioxide (21).

Table 3 shows the flue gas mass and volume flows entering the flue gas treatment pass and the flue gas recycle loop in the $O_2/CO_2$ power plant. Hence, wet recycling conditions are applied in the present process layout. This is accompanied by an increased level of $SO_2$ and other impurities in the boiler. However, the advantage of this recycle approach is that the flue gas flow entering the cleaning equipment is drastically reduced, which leads to savings in the investment costs. Given that the FGD is included in the process, no drastic changes on the overall design of the flue gas treatment would be required if, for fuel specific reasons mainly, dry and clean flue gas recycling conditions need to be applied. One such reason may be high concentrations of $SO_3$, causing the sulphuric acid dew point to drop. In principle, the only difference is that the recycle line would need to be changed to include the flue gas condenser and an FGD ((13) and (14) in Fig. 2) and that the size of these two units must be increased to enable treatment of the total flue gas flow.

Various options for utilization of the available flue gas heat, more than 500 MW, downstream of the economizer (11) in the capture plant have been evaluated. The heat available at a minimum temperature of 90 °C could be used in an absorption cooler to produce a coolant stream at 5 °C, e.g. to be used for sub-cooling purposes of the $CO_2$ or cooling of the compressors. Especially, the large amount of heat rejected from the flue gas condenser (13), approximately 365 MW between 88 °C and 60 °C where most of the heat of evaporation is released, will require a heat sink. Part of this heat could be used in the available district heating system, but the one alternative represented in the results of this work is the integration with the steam cycle to preheat the feed water in order to increase the electricity output of the plant instead of using low pressure steam. The details on these calculations are given elsewhere [9,10].

The flue gas treatment basically involves the removal of water and non-condensable gases. Fig. 3 shows the mass flow of the flue gas components throughout the treatment steps, considering the two options mentioned above with respect to excluding and including an FGD. Both options are provided, since $SO_2$ may cause problems depending on the type of storage considered. For example, storing the $SO_2$ below ground level can cause problems due to its sulfation behaviour in contact with the calcium present in the storage environment. If sulphates are produced, e.g. in a saline aquiphere, the porosity will decrease, which will directly affect the storage capacity.

Besides the technical feasibility, a combined underground storage of $CO_2$ and $SO_2$ also depends on political decisions with respect to dumping conventions. A complete dehydration of the flue gas is important, since it will reduce the mass flow and inhibit corrosion and hydrate precipitation. In the case of combined storage of $CO_2$ and $SO_2$, provided the flue gas is dehydrated to a dew point 5° below the temperature required for transport conditions, the sulphur dioxide will, in principle, behave as the carbon dioxide, and the two gases will not cause any corrosion problems. For both storage options, the gas must, nevertheless, be dehydrated before reaching the high pressure steps in the compression to make the compression of the gas mixture possible [23].

Carbon dioxide alone can be corrosive in the presence of water and cause the so called sweet corrosion, i.e. when water vapour in the gas form solid ice like crystals called gas hydrates. The hydrates are formed when the water encages gas molecules smaller than 1.0 nm (which is the case for both carbon dioxide and sulphur dioxide) at low temperatures and elevated pressures (below 25 °C and above 15 bar). Various mechanisms for the carbon dioxide corrosion process have been proposed, all of which involve either carbonic acid or bicarbonate ion formed when the carbon dioxide is dissolved in water. Also, in this case, dehydration to a dew point five degrees below the transport temperature is sufficient to avoid the problem [24], since dry carbon dioxide is not corrosive at temperatures below 400 °C [25]. The maximum water content in the gas prior to compression should, therefore, not exceed 60–100 mg/$m_n^3$ [26], whatever the content of other possible acidic compounds. For pipeline transportation with the presence of water, serious corrosion can be expected if the partial pressure of carbon dioxide exceeds 2 bar [27].

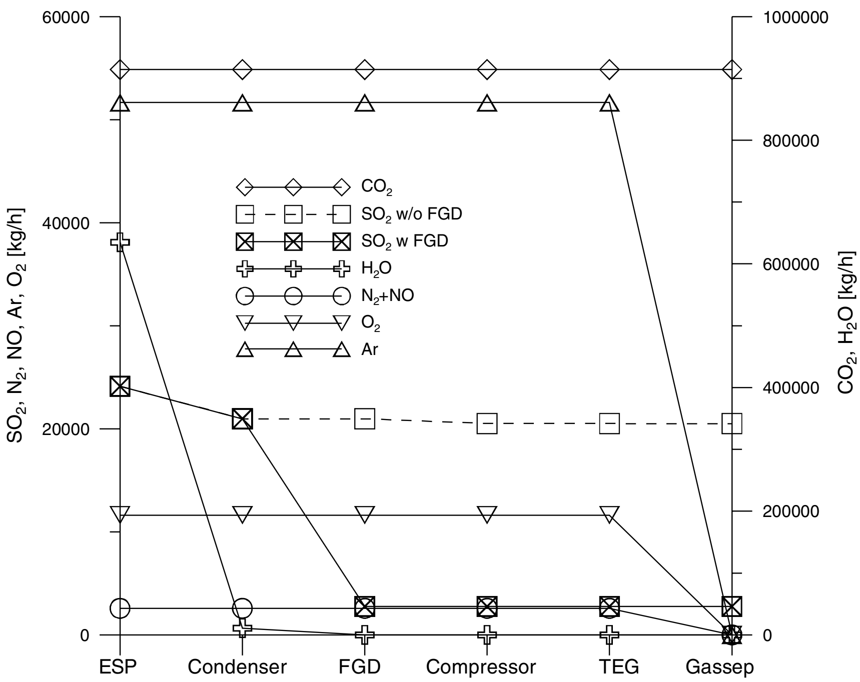



Fig. 3. Flue gas mass flows [kg/h] through the flue gas treatment steps.

In the process chosen, the gas is dehydrated in two steps. The first is in a traditional flue gas condenser (13) where most of the water is removed, together with the remaining particles ($SO_3$, etc.). The second dehydration step is the triethylene glycol (TEG) unit (16), which will remove the remaining water down to a value of 60 mg/$m_n^3$, corresponding to a dew point of −5 °C at 100 bar under transport conditions. Since the TEG requires a pressure of 30 bar to be efficient, a compressor step with intercooling is installed before the TEG. Some water is separated in the cooling steps of the compressor.

To reduce the power consumption of the flue gas compressors, compression up to the transport pressure is performed by a high pressure pump (22). This is because the pressure should only be increased to a sufficient level to allow transition of the flue gas (mostly carbon dioxide) into a liquid state at a reasonable cost. The first compressor step raises the pressure from 1 bar to 30 bar, which is the inlet pressure of the TEG. The flue gas is then compressed in the second step from 30 bar to 58 bar. At a pressure of 58 bar, the carbon dioxide and sulphur dioxide will be liquefied, if cooled to 20 °C by the main cooling system. When the carbon dioxide is liquefied, a high pressure pump is used for the last pressure increase up to 100 bar before transportation to the injection site.

### 3.2. Plant efficiency and emissions

Fig. 4 shows a Sankey diagram for the reference power plant (A) and the $O_2/CO_2$ power plant (B). The figure illustrates the energy losses in the $O_2/CO_2$ plant with the same net electricity output as the reference plant. The net electrical efficiency of the plant becomes 33.5%, which is to be compared with 42.6% for the reference plant, i.e. the energy penalty of the capture plant, according to Eq. (2), becomes 21.5%. The plant in Fig. 4 represents the option including $SO_2$ removal. In the case of the combined capture of $SO_2$ and $CO_2$, the internal electricity demand is reduced, with a boiler power of 2524 MW, and the net efficiency is slightly increased to 34.3% (see the power plant specifications in Table 6). The Sankey diagram of the capture plant includes the auxiliary benefits from a reduced flue gas flow as well as the integration possibilities as discussed

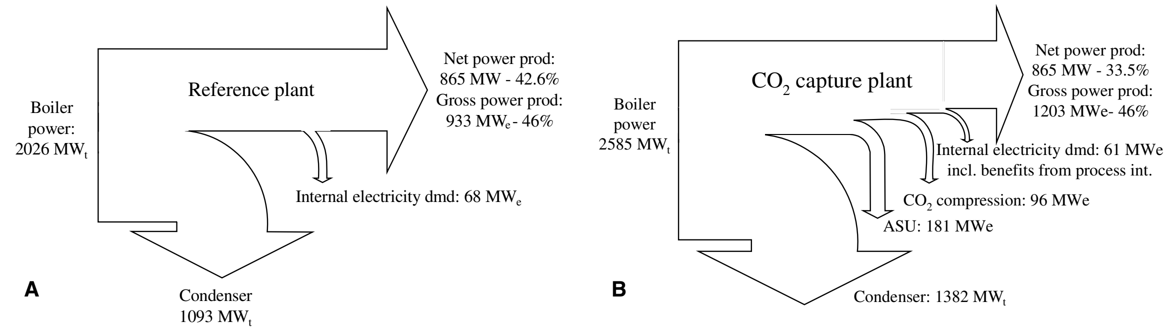



Fig. 4. Sankey diagrams of the reference power plant (A) and the $CO_2$ capture plant (B). The capture plant has the same power output as the reference power plant, scaled according to Eq. (1).

Table 4
Comparison of emissions to atmosphere between the reference plant and the $O_2/CO_2$ power plant

| Emissions to air | Reference plant | | | $O_2/CO_2$-plant | | |
|---|---|---|---|---|---|---|
| | [mg/$m_n^3$] | [kg/h] | [kg/MWh$_e$] | [mg/$m_n^3$] | [kg/h] | [kg/MWh$_e$] |
| $SO_x$ | <350 | <1120 | 1.28 | <6 | <13 | 0.015 |
| $NO_x$ | <145 | <460 | 0.53 | <141 | <190 | 0.220 |
| $CO_2$ | <235 | <740,000 | 855.2 | <4 | <5000 | 5.8 |
| Dust | <2 | <6 | 0.007 | <1 | <1 | 0.001 |

previously. The electricity production is increased by about 10 MW due to feed water preheating with the heat from the flue gas condensation.

Table 4 summarizes emissions to the atmosphere from the $O_2/CO_2$ fired plant obtained in the present study and compares these with those of the reference plant. The combustion conditions are considered to be stoichiometric with an oxygen excess of 1.5% on a dry basis. The $SO_x$, $NO_x$ and $CO_2$ emissions are leakage flows and ventilated flows from the condensation unit (13), the TEG unit (16) and the separator (20) in the flue gas treatment pass.

Estimation of $NO_x$ formation is based on the results in [7] with a reduction of the emissions of about 60%. The reduction can be attributed to the absence of thermal $NO_x$ as well as a drastic reduction in the conversion ratio of fuel nitrogen to exhaust $NO_x$ as reported by [8,28]. The emitted $NO_x$ is ventilated to the air in a concentrated stream in the gas/liquid separator (20) since it is assumed to be non-soluble in the $CO_2/SO_2$ mixture. In order to reach further reduction in $NO_x$ emissions, the high $NO_x$ concentration stream from separator (20) should be well suited for a $NO_x$ catalyst.

### *3.3. Economic evaluation*

Table 5 lists the overall cost parameters used together with the capture and avoidance costs according to Eq. (1) and the definitions described above. In Table 5, an interest rate of 10% has been assumed as a mid-range value compared to previously preformed studies on the economics of $CO_2$ capture [1–4] where interest rates between 7% and 15% have been assumed. This is also in line with the standard power plant economic and assessment criteria introduced by IEA [29], which suggests an interest rate set at 10% and an assumed load factor of 85%. However, according to OECD [21], the long-term interest rate (10-year basis) is forecasted to be around 4% in the Euro area (March, 2005), Fig. 5 gives the $CO_2$ avoidance cost for different interest rates and fuel prices, and obviously, these parameters have a significant impact on the results. It should, hence, be noted that this figure is likely to be lower in European projects and in a previous paper by the authors [30] where an interest rate of 6% was applied. The lignite price, economic life time of the plant and distribution of costs during construction are based on information from industry.

The energy availability of the plant is 7500 h/year at full capacity, which corresponds to a load factor of 85%. As is seen in Table 6, the electricity generation cost increases from 42.1 to 64.3 $/MWh, which corresponds to a $CO_2$ avoidance cost of 26 $/ton $CO_2$ (or 20 €/ton $CO_2$). Table 6 also shows that without FGD, the avoidance cost decreases about 4 $/ton (3 €/ton $CO_2$).

As previously discussed, storage of $CO_2$ contaminated with $SO_2$ may be difficult from both a legal and public acceptance point of view. Still, the results show that combined storage has a marginal effect on the overall cost situation. It should be pointed out that the resulting cost is strongly dependent on the economical parameters for the annuity cost calculations (although this should be rather obvious). As shown in Fig. 5,

Table 5
Input cost parameters

| Cost parameters | | |
|---|---|---|
| O&M cost assumptions | Power plant—variable [€/MWh] | 1.0 |
| | Power plant—fixed [%] | 1.5 |
| | ASU—fixed [%] | 4.0 |
| | Flue gas treatment—fixed [%] | 4.0 |
| Distribution of costs over construction period: 4 years | 1st year | 0.15 |
| | 2nd year | 0.30 |
| | 3rd year | 0.35 |
| | 4th year | 0.20 |
| Interest rate [%] | | 10.0 |
| Lignite price [€/MWh] | | 4.00 |
| Exchange rate [€/$] | | 1.30 |
| Plant economic life time [years] | | 20 |

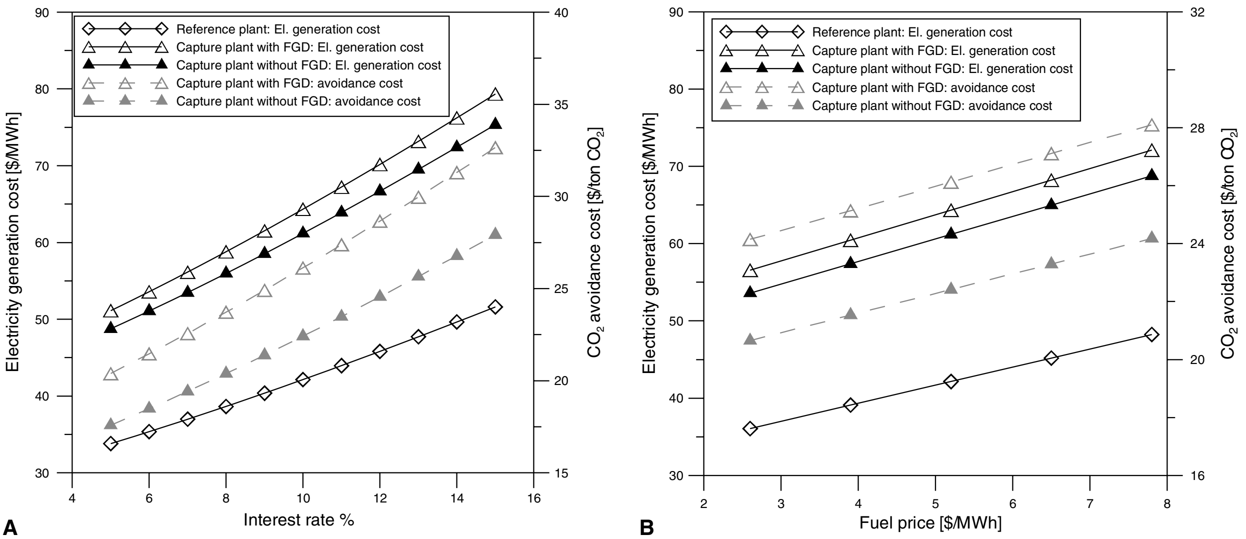



Fig. 5. Avoidance cost as a function of: (A) interest rate on invested capital (lignite price as in Table 5); and (B) fuel price (interest rate as in Table 5).

Table 6
Plant performance and costs of the reference plant and the $O_2/CO_2$ power plant with and without FGD (the cost of the FGD is included in the power plant cost)

| Plant performance and costs | | Reference | $O_2/CO_2$ w FGD | $O_2/CO_2$ w/o FGD |
|---|---|---|---|---|
| Boiler power [MW] | | 2026 | 2585 | 2524 |
| Gross electrical output [MW] | | 933 | 1203 | 1176 |
| Net electrical output [MW] | | 865 | 865 | 865 |
| Operating time [h] | | 7500 | 7500 | 7500 |
| Fuel demand [kg/s] | | 192.9 | 240.4 | 246.2 |
| $O_2$ demand [kg/s] | | – | 221.6 | 226.9 |
| Total investment costs [$10^6 \times$ €] | Power plant | 1100 | 1403.50 | 1310.74 |
| | ASU | – | 304.59 | 297.06 |
| | Flue gas treatment | – | 35.46 | 34.70 |
| Annualized capital cost [$10^6 \times$ €/year] | | 126.55 | 200.59 | 188.97 |
| O&M costs: fixed + variable [$10^6 \times$ €/year] | Power plant | 22.99 | 29.33 | 27.39 |
| | ASU | – | 12.18 | 11.88 |
| | Flue gas treatment | – | 1.42 | 1.39 |
| Annualized O&M cost [$10^6 \times$ €/year] | | 22.99 | 42.93 | 40.66 |
| Fuel costs [$10^6 \times$ €/year] | | 60.77 | 77.55 | 75.72 |
| Total annualized costs [$10^6 \times$ €/year] | | 210.31 | 321.07 | 305.35 |
| Electricity generation cost [€/MWh] | | 32.4 | 49.5 | 47.1 |
| Electricity generation cost [$/MWh] | | 42.1 | 64.3 | 61.2 |
| Emitted $CO_2$ [ton/MWh] | | 0.855 | 0.0058 | 0.0058 |
| Avoidance cost [€/ton $CO_2$] | | – | 20.0 | 17.1 |
| Avoidance cost [$/ton $CO_2$] | | – | 26.0 | 22.3 |

the avoidance cost differs substantially, depending both on the interest rate chosen for the invested capital (Fig. 5A) and on the fuel cost (Fig. 5B). Thus, the latter should be kept in mind when comparing this study with similar studies given in the literature.

## 4. Conclusions

This study proposes an overall process scheme of an $O_2/CO_2$ plant (Fig. 2) based on commercial data for the key components required in the process. With all integration possibilities considered, the net efficiency becomes approximately 33.5%, which should be compared to 42.6% in the reference plant. An almost complete dehydration of the flue gas is of great importance to avoid problems in the final flue gas treatment and in the transportation of the carbon dioxide. The fixed and running costs associated with an 865 $MW_e$ lignite fired $O_2/CO_2$ power plant have been evaluated in order to obtain the $CO_2$ avoidance cost for a new state of the art capture plant. The capture plant has been scaled up to yield the same net capacity as the reference plant (Fig. 4B), and the $CO_2$ emissions to the atmosphere are reduced 99.5%. With a lignite price of 5.2 $/MWh (4.0 €/MWh) and an interest rate of 10%, the electricity generation cost increases from 42.1 to 64.3 $/MWh, which corresponds to a $CO_2$ avoidance cost of 26 $/ton $CO_2$ (or 20 €/ton $CO_2$). Furthermore, the study shows that if combined capture and storage of $CO_2$ and $SO_2$ is environmentally approved and applicable to the type of storage considered, the economic benefit for the plant studied is still small, with a reduction in the electricity generation cost of 3.1 $/MWh. In summary, by using commercial data from existing plants and components, this study shows that $O_2/CO_2$ combustion is a realistic and a near future option for $CO_2$ reductions in the power sector.

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